



Closing the loop, not just one experiment.

A Design → Make → Test → Analyze loop for TEM-1 β -lactamase screening; agent-run, scientist-verified.

DMTA

design, make, test,
analyze, closed loop

2 New assays

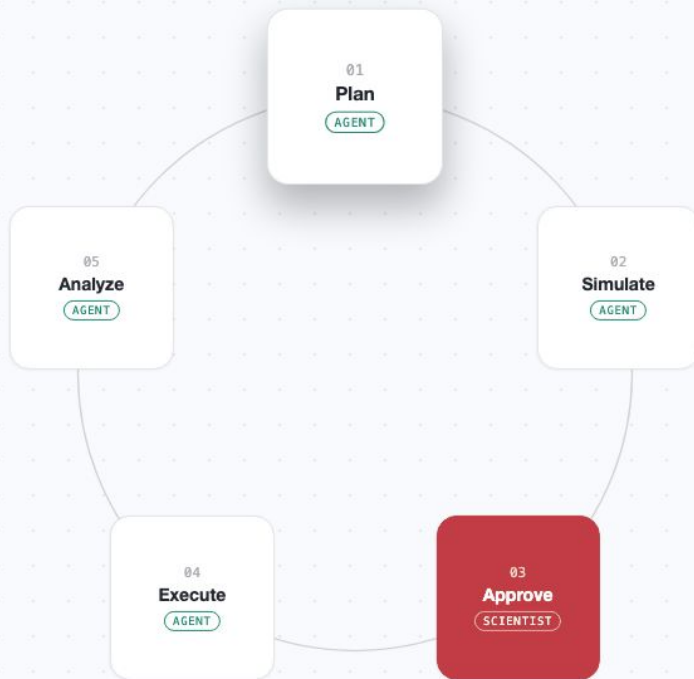
new workflows, one
shared skill library

Agentic Optimization

120 actions, 33 min → 19 min

Agent-Driven Discovery Loop

Plan → Simulate → Approve → Execute → Analyze — TEM-1 β -lactamase inhibitor screening



01 Plan

AGENT

Selects run conditions, checks every leg against the pipette's real range.

02 Simulate

AGENT

Deterministic validation — values sync into the workflow the app actually opens.

03 Approve

SCIENTIST

Human reviews the plan and simulation. The agent cannot start a run.

04 Execute

AGENT

Physical robot arm runs it — tip-indexed and logged, start to finish.

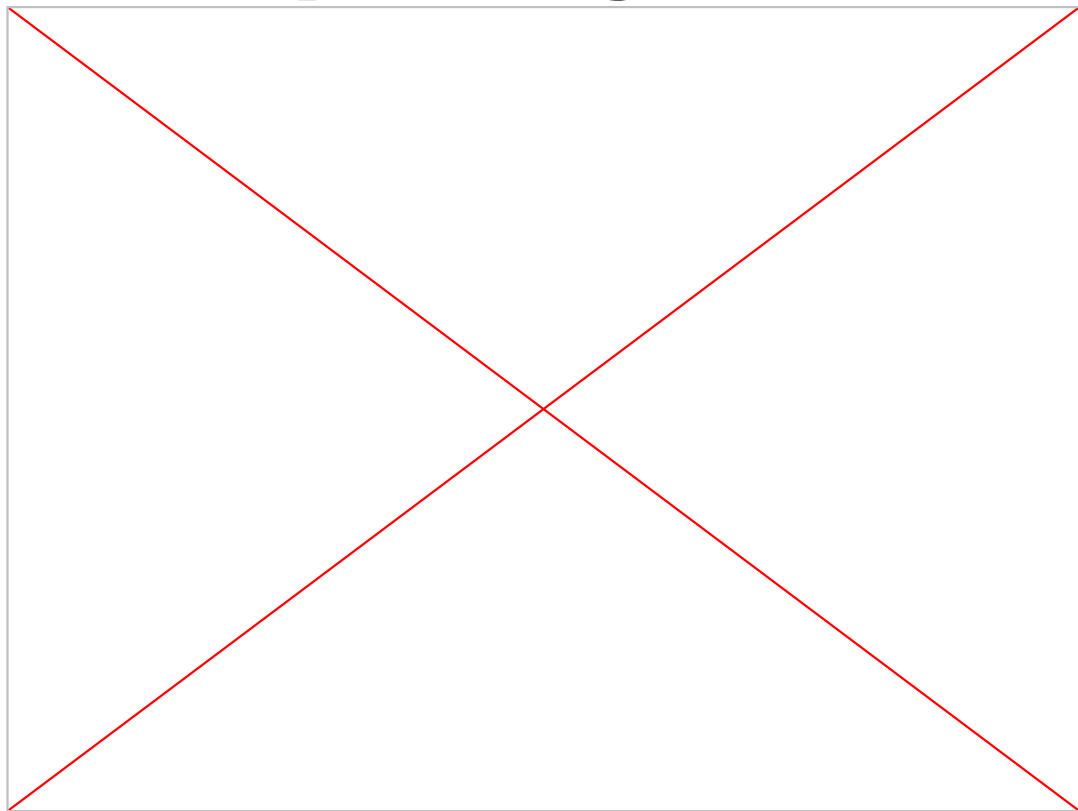
05 Analyze

AGENT

Reads the plate-reader export, flags pass/fail, feeds the next round.



The loop, running.

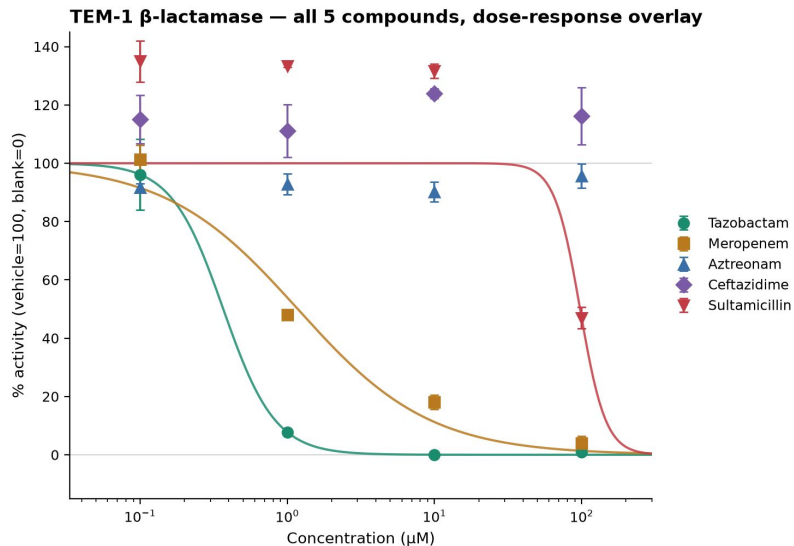


- 01** Design: agent proposes conditions; scientist confirms they're rigorous
- 02** Make: the robot builds the reaction
- 03** Test — the robot reads the result
- 04** Analyze: agent recommends; scientist verifies before the next Design

RESULTS



The loop ran end-to-end. One step still needs the right context.



Tazobactam 0.367µM • Meropenem 1.18µM

Aztreonam / Ceftazidime: no inhibition to 100µM

Sultamicillin: 97.6µM, CI unresolved — flagged, not confirmed

Next

- Screen the other 15 top-20 compounds — same validated assay
- Resolve Sultamicillin — real hit or 100µM artifact?
- Tighten Meropenem's ladder — anchored vs. free fit disagree 3×

30 min for 11 wells → 20 min for 48 wells



The plumbing is done. Scientific discovery awaits.

Real today

- Agentic planning, workflow generation, data setup, data analysis, data curation
- Full history in beads & visualized in marimo
- Analyst already flags weak fits, suggests a better range
- Agent proposes, scientist triggers
- Cost calculation for agent usage

Next

- Asking agents to create more skills
- Agentic kick off of runs
- Cross-round compound selection

The scientist stays the rigor gate. Plan, run, read, and recommend are the pipeline's job.

THANK YOU



Thank you.

Zeon Systems

the hardware, platform and the opportunity

Sepia Bio

the CFPS kits and hands on expertise behind every well

Our judges

Ale, Isaac, Keltoum, Jimmy & Haran for the time and the questions

Team MIKMAK — AI for Science World Models Hack